5.3.1 The principles and importance of homeostasis

	Learning outcomes	Additional guidance
	Learners should be able to demonstrate and apply their knowledge and understanding of:	
(a)	the general principles of homeostasis in the maintenance of a stable internal environment	To include the role of receptors, effectors and negative feedback **AND** the concept of normal ranges for temperature, pH, blood glucose concentration and blood pressure. **AND** contrast positive and negative feedback systems using oxytocin as an example of positive feedback. *M1.6*
(b)	the nervous and hormonal control of heart rate	To include the roles of the sympathetic and parasympathetic nervous system and adrenaline.
(c)	the control of body temperature	To include the role of peripheral temperature receptors, the thermoregulatory centre in the hypothalamus and the responses to rising and falling temperatures (sweating, shivering, vasoconstriction and vasodilation).
(d)	the regulation of thyroxine release and the effect of thyroxine on metabolic rate	
(e)	the techniques for and the importance of measuring core body temperature	To include oral, tympanic (ear), axillary (arm pit) and rectal methods and the significance of readings outside the normal range in adults and children. *M1.1, M1.2, M1.6* **PAG10** HSW3, HSW4, HSW5, HSW6

(f)	the causes, symptoms and treatment of hypothermia and hyperthermia.	To include the consequences of 'fuel poverty' and analysis of data showing correlations between weather conditions and incidence of both hypothermia and hyperthermia with respect to climate change. *M0.3, M1.7* HSW9, HSW10, HSW11, HSW12